twiners (*Tragia involucrata*) some are xerophytic (*Euphorbia*) with cactus – like (phylloclades) plants usually contain milky or watery sap.

Root: Well branched tap root system.

Stem: Aerial, erect or prostrate (*E.prostrata*), herbaceous or woody. Stem becomes modified into flattened, leaf-like and becomes succulent in several species of *Euphorbia.* Such modified stem is called phylloclades. Cylindrical, branched, solid or hollow, usually contain latex either milky (*E.tirucalli*) or watery (*Jatropha curcas*).

Leaf: Stipulate or exstipulate. Mostly simple, alternate, often reduced or deciduous as in several species of *Euphorbia*, palmately lobed in (*Ricinus)* or deeply lobed in (*Manihot)*. The stiplues are modified into a pair of spines (*E.splendens*) or glandular hairs (*Jatropha curcas*). The leaves around the cyathium inflorescence become beautifully coloured in *E.pulcherrima* (Paalperukki tree) with unicostate or multicostate reticulate venation.

Inflorescence: The inflorescence of Euphorbiaceae varies greatly, Terminal compound raceme – *Croton, Ricinus,* Catkin – *Acalypha hispida* Cyme - *Jatropha,* solitary axillary – *Phyllanthus asperulatus,* cyathium – *Euphorbia* species.

Cyathium is an unique and special inflorescence of this family.

Flowers: Bracteate, ebracteolate, generally unisexual, homochlamydeous, rarely heterochlamydeous, monoecious or dioecious, actinomorphic, rarely zygomorphic, hypogynous, rearly perigynous.

Perianth: Tepals 0 to 5 **biseriate** (male flowers of *Croton bonplandianum*) uniseriate or aphyllous (*Euphorbia*), valvate or imbricate when present, apophyllous or synphyllous.

Androecium: Stamens are 1 to many. In *Euphorbia* a single stalked stamen represents a single male flower. In *Ricinus,* usually 5 stamens are present, but each stamen is profusely branched. In *Jatroph*a, they are arranged in two whorls each of 5 stamens. The anthers are dithecous, dehisce either by transverse or longitudinal slits.

Gynoecium: Tricarpellary, rarely bicarpellary(*Bridelia*),tetraorpentacarpellary (*Wielandia*), syncarpous, ovary superior, rarely semi-inferior, ovules one or two in each locule on axile placentation, Styles 3, each split into two feathery stigma. Nectaries are usually present, gynoecium is present as a pistillode in staminate flowers.

Fruit: Fruits are capsule or schizocarp. It breaks violently and dehisce into three one seeded cocci called regma (*Ricinus*), drupe in (*Emblica officinalis)* and berry or samara.

Seed: Seeds are endospermous. In *Ricinus* knob-like caruncle develops from the micropyle,that absorbs and temporarily retains water enabling germination.

Botanical description of *Ricinus communis* (Castor)

Habit: Tall perennial shrub

Root: Branched tap root system

Stem: Aerial, erect, cylindrical, branched and hollow, solid at the base, glabrous,

Leaf: Simple, petiolate, hollow, exstipulate, alternate, broad, palmately lobed, usually 7-9 lobes, serrate, palmately reticulate divergent venation.

Inflorescence: Terminal panicle.

Male Flower Bracteate, ebracteolate, pedicellate, male flowers (open for one day) towards lower portion of the inflorescence, actinomorphic, incomplete.

Perianth: Tepals 5, apophyllous, uniseriate, green, valvate aestivation, odd tepal posterior in position.

Androecium: Stamens numerous each stamen profusely branched, anthers globose basifixed.

Gynocium: usually absent rarely represented by pistillode.

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